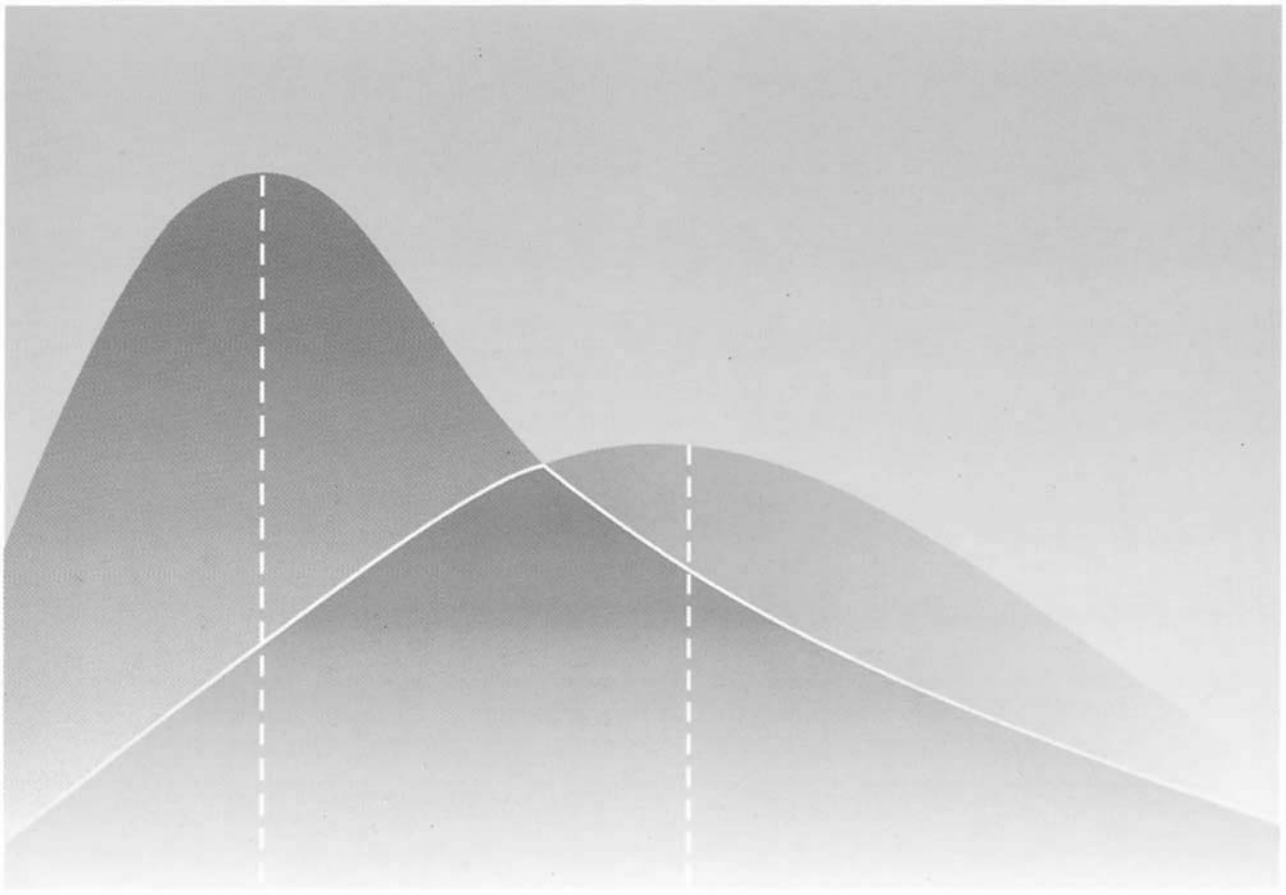
Probabilistic Methods of Signal and System Analysis

THIRD EDITION



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6–9 Correlation Matrices for Sampled Functions

The discussion of correlation thus far has concentrated on only two random variables. Thus, for stationary processes the correlation functions can be expressed as a function of the single variable τ. There are many practical situations, however, in which there may be many random variables and it is necessary to develop some convenient method for representing the many autocorrelations and crosscorrelations that arise. The use of vector notation provides a convenient way of representing a set of random variables, and the product of vectors that is necessary to obtain correlations results in a matrix. It is important, therefore, to discuss some situations in which the vector representation is useful and to describe some of the properties of the resulting correlation matrices. A situation in which vector notation is useful in representing a signal arises in the case of a single time function that is sampled at periodic time instants. If only a finite number of such samples are to be considered, say N, then each sample value can become a component of an $(N \times 1)$ vector. Thus, if the sampling times are $t_1, t_2, \ldots, t_N$, the vector representing the time function $X(t)$ may be expressed as

$$\boldsymbol{X} = \begin{bmatrix} X(t_1) \\ X(t_2) \\ \vdots \\ X(t_N) \end{bmatrix}$$

If $X(t)$ is a sample function from a random process, then each of the components of the vector $\boldsymbol{X}$ is a random variable.

It is now possible to define a correlation matrix that is $(N \times N)$ and gives the correlation between every pair of random variables. Thus,

$$\boldsymbol{R}_X = E[\boldsymbol{X}\boldsymbol{X}^T] = E\begin{bmatrix} X(t_1)X(t_1) & X(t_1)X(t_2) & & X(t_1)X(t_N) \\ X(t_2)X(t_1) & X(t_2)X(t_2) & & \\ \vdots & & & \\ X(t_N)X(t_1) & & & X(t_N)X(t_N) \end{bmatrix}$$

where $\boldsymbol{X}^T$ is the transpose of $\boldsymbol{X}$. When the expected value of each element of the matrix is taken, that element becomes a particular value of the autocorrelation function of the random process from which $X(t)$ came. Thus.

$$\boldsymbol{R}_X = \begin{bmatrix} R_X(t_1, t_1) & R_X(t_1, t_2) & & R_X(t_1, t_N) \\ R_X(t_2, t_1) & R_X(t_2, t_2) & & \\ \vdots & & & \\ R_X(t_N, t_1) & & & R_X(t_N, t_N) \end{bmatrix} \tag{6–45}$$

When the random process from which $X(t)$ came is wide-sense stationary, then all the components of $\boldsymbol{R}_X$ become functions of time difference only. If the interval between sample values is Δt, then

$$t_2 = t_1 + \Delta t$$

$$t_3 = t_1 + 2\Delta t$$

$$t_N = t_1 + (N-1)\Delta t$$

and

$$\boldsymbol{R}_X = \begin{bmatrix} R_X[0] & R_X[\Delta t] & \cdots & R_X[(N-1)\Delta t] \\ R_X[\Delta t] & R_X[0] & & \\ \vdots & & & \\ R_X[(N-1)\Delta t] & & & R_X[0] \end{bmatrix} \tag{6–46}$$

where use has been made of the symmetry of the autocorrelation function; that is, $R_X[i\,\Delta t] = R_X[-i\,\Delta t]$. Note that as a consequence of the symmetry, $\boldsymbol{R}_X$ is a symmetric matrix (even in the nonstationary case), and that as a consequence of stationarity, the major diagonal (and all diagonals parallel to it) have identical elements.

Although the $\boldsymbol{R}_X$ just defined is a logical consequence of previous definitions, it is not the most customary way of designating the correlation matrix of a random vector consisting of sample values. A more common procedure is to define a *covariance matrix*, which contains the variances and covariances of the random variables. The general covariance between two random variables is defined as

$$E\{[X(t_i) - \overline{X}(t_i)][X(t_j) - \overline{X}(t_j)]\} = \sigma_i \sigma_j \rho_{ij} \tag{6–47}$$

where $\overline{X}(t_i)$ = mean value of $X(t_i)$
$\overline{X}(t_j)$ = mean value of $X(t_j)$
σ_i^2 = variance of $X(t_i)$
σ_j^2 = variance of $X(t_j)$
ρ_{ij} = normalized covariance coefficient of $X(t_i)$ and $X(t_j)$
= 1, when $i = j$

The covariance matrix is defined as

$$\boldsymbol{\Lambda}_X = E[\boldsymbol{X} - \overline{\boldsymbol{X}})(\boldsymbol{X}^T - \overline{\boldsymbol{X}}^T)] \tag{6–48}$$

where $\overline{\boldsymbol{X}}$ is the mean value of $\boldsymbol{X}$. Using the covariance definitions leads immediately to

$$\Lambda_X = \begin{bmatrix} \sigma_1^2 & \sigma_1\sigma_2\rho_{12} & & \sigma_1\sigma_N\rho_{1N} \\ \sigma_2\sigma_1\rho_{21} & \sigma_2^2 & & \\ \vdots & & & \\ \sigma_N\sigma_1\rho_{N1} & & & \sigma_N^2 \end{bmatrix} \tag{6-49}$$

since $\rho_{ii} = 1$, for $i = 1, 2, \ldots, N$. By expanding (6–49) it is easy to show that Λ_X is related to R_X by

$$\Lambda_X = R_X - \overline{X}\,\overline{X}^T \tag{6-50}$$

If the random process has a zero mean, then $\Lambda_X = R_X$.

The above representation for the covariance matrix is valid for both stationary and nonstationary processes. In the case of a wide-sense stationary process, however, all the variances are the same and the correlation coefficients in a given diagonal are the same. Thus,

$$\sigma_i^2 = \sigma_j^2 = \sigma^2 \qquad i, j = 1, 2, \ldots, N$$

$$\rho_{ij} = \rho_{|i-j|} \qquad i, j = 1, 2, \ldots, N$$

and

$$\Lambda_X = \sigma^2 \begin{bmatrix} 1 & \rho_1 & \rho_2 & & \rho_{N-1} \\ \rho_1 & 1 & \rho_1 & \cdots & \rho_{N-2} \\ \rho_2 & \rho_1 & 1 & \rho_1 & \\ & \cdots & & & \\ & & & & \\ & & & & \\ & & & & 1 \quad \rho_1 \\ \rho_{N-1} & & & & \rho_1 \quad 1 \end{bmatrix} \tag{6-51}$$

Such a matrix is said to be *Toeplitz*.

As an illustration of some of the above concepts, suppose we have a stationary random process whose autocorrelation function is given by

$$R_X(\tau) = 10e^{-|\tau|} + 9 \tag{6-52}$$

To keep the example simple, assume that three random variables separated by 1 second are to be considered. Thus, $N = 3$ and $\Delta t = 1$. Evaluating (6–52) for $\tau = 0, 1, 2$ yields the values that are needed for the correlation matrix. Thus, the correlation matrix becomes

$$R_X = \begin{bmatrix} 19 & 12.68 & 10.35 \\ 12.68 & 19 & 12.68 \\ 10.35 & 12.68 & 19 \end{bmatrix}$$

Since the variance of this process is 10 and its mean value is ± 3, the covariance matrix is

$$\mathbf{\Lambda}_X = 10\begin{bmatrix} 1 & 0.368 & 0.135 \\ 0.368 & 1 & 0.368 \\ 0.135 & 0.368 & 1 \end{bmatrix}$$

Another situation in which the use of vector notation is convenient arises when the random variables come from different random processes. In this case, the vector representing all the random variables might be written as

$$\boldsymbol{X}(t) = \begin{bmatrix} X_1(t) \\ X_2(t) \\ \vdots \\ X_N(t) \end{bmatrix}$$

The correlation matrix is now defined as

$$\boldsymbol{R}_X(\tau) = E[\boldsymbol{X}(t)\boldsymbol{X}^T(t+\tau)] \tag{6–53}$$

$$= \begin{bmatrix} R_1(\tau) & R_{12}(\tau) & \cdots & R_{1N}(\tau) \\ R_{21}(\tau) & R_2(\tau) & & \\ \vdots & & & \\ R_{N1}(\tau) & & & R_N(\tau) \end{bmatrix}$$

in which

$$R_i(\tau) = E[X_i(t)X_i(t+\tau)]$$

$$R_{ij}(\tau) = E[X_i(t)X_j(t+\tau)]$$

Note that in this case, the elements of the correlation matrix are functions of τ rather than numbers as they were in the case of the correlation matrix associated with samples taken from a single random process. Situations in which such a correlation matrix might occur arise in connection with antenna arrays or arrays of seismic detectors. In such systems, the noise signals at each antenna element, or each seismic detector, may be from different, but correlated, random processes.

Before we leave the subject of covariance matrices, it is worth noting the important role that these matrices play in connection with the joint probability density function for N random variables from a Gaussian process. It was noted earlier that the Gaussian process was one of the few for which it is possible to write a joint probability density function for any number of random variables. The derivation of this joint density function is beyond the scope of this discussion, but it can be shown that it becomes

$$f(\boldsymbol{x}) = f[x(t_1), x(t_2), \ldots, x(t_N)]$$

$$= \frac{1}{(2\pi)^{N/2}\,|\mathbf{\Lambda}_X|^{1/2}} \exp\left[-\frac{1}{2}(\boldsymbol{x}^T - \overline{\boldsymbol{x}}^T)\mathbf{\Lambda}_x^{-1}(\boldsymbol{x} - \overline{\boldsymbol{x}})\right] \tag{6–54}$$

where $|\mathbf{\Lambda}|_X$ is the determinant of $\mathbf{\Lambda}_X$ and $\mathbf{\Lambda}_X^{-1}$ is its inverse.

The concept of correlation matrices can also be extended to represent crosscorrelation functions. Suppose we have two random vectors $\boldsymbol{X}(t)$ and $\boldsymbol{Y}(t)$ where each vector contains N random variables. Thus, let

$$\boldsymbol{X}(t) = \begin{bmatrix} X_1(t) \\ X_2(t) \\ \vdots \\ X_N(t) \end{bmatrix} \qquad \boldsymbol{Y}(t) = \begin{bmatrix} Y_1(t) \\ Y_2(t) \\ \vdots \\ Y_N(t) \end{bmatrix}$$

By analogy to (6–53) the crosscorrelation matrix can be defined as

$$\begin{aligned} \boldsymbol{R}_{XY}(\tau) &= E[\boldsymbol{X}(t)\boldsymbol{Y}^T(t+\tau)] \\ &= \begin{bmatrix} R_{11}(\tau) & R_{12}(\tau) & & R_{1N}(\tau) \\ R_{21}(\tau) & R_{22}(\tau) & & \\ \vdots & & & \\ R_{N1}(\tau) & & & R_{NN}(\tau) \end{bmatrix} \end{aligned} \tag{6–55}$$

where now

$$R_{ii}(\tau) = E[X_i(t)Y_i(t+\tau)]$$

$$R_{ij}(\tau) = E[X_i(t)Y_j(t+\tau)]$$

In many situations the vector of random processes $\boldsymbol{Y}(t)$ is the sum of the vector $\boldsymbol{X}(t)$ and a statistically independent noise vector $\boldsymbol{N}(t)$ that has zero mean. In this case, (6–55) reduces to the autocorrelation matrix of (6–53) because the crosscorrelation between $\boldsymbol{X}(t)$ and $\boldsymbol{N}(t)$ is identically zero. There are other situations in which the elements of the vector $\boldsymbol{Y}(t)$ are time delayed versions of a single random process $X(t)$. Also unlike the autocorrelation matrix of (6–53), it is not necessary that $\boldsymbol{X}(t)$ and $\boldsymbol{Y}(t)$ have the same number of dimensions. If $\boldsymbol{X}(t)$ is a column vector of size M and $\boldsymbol{Y}(t)$ is a column vector of size N, the crosscorrelation matrix will be an $M \times N$ matrix instead of a square matrix. This type of matrix may arise if $\boldsymbol{X}(t)$ is the single wideband random input to a system and the vector $\boldsymbol{Y}(t)$ is composed of responses at various points in the system. As discussed further in a subsequent chapter, the crosscorrelation matrix, which is now a $1 \times N$ row vector, can be interpreted as the set of impulse responses at these various points.

Exercise 6–9.1

A random process has an autocorrelation function of the form

$$R_X(\tau) = 10e^{-|\tau|} \cos 2\pi$$

Write the correlation matrix associated with four random variables defined for time instants separated by 0.5 second.

Answers: Elements in the first row include 3.677, 2.228, 10.0, 6.064

Exercise 6–9.2

A covariance matrix for a stationary random process has the form

$$\begin{bmatrix} 1 & 0.6 & 0.4 & — \\ — & 1 & 0.6 & — \\ 0.4 & 0.6 & — & 0.6 \\ 0.2 & — & — & 1 \end{bmatrix}$$

Fill in the blank spaces in this matrix.

Answers: 1, 0.6, 0.2, 0.4

PROBLEMS

6–1.1 A stationary random process having sample functions of $X(t)$ has an autocorrelation function of

$$R_X(\tau) = 5e^{-5|\tau|}$$

Another random process has sample functions of

$$Y(t) = X(t) + bX(t - 0.1)$$

a) Find the value of b that minimizes the mean-square value of $Y(t)$.

b) Find the value of the minimum mean-square value of $Y(t)$.

c) If $|b| \leq 1$, find the maximum mean-square value of $Y(t)$.

6–1.2 For each of the autocorrelation functions given below, state whether the process if represents might be wide-sense stationary or cannot be widesense stationary.

a) $R_X(t_1, t_2) = e^{t_1}e^{-t_2}$